

Design Assignment - 2

→ Design a 2-stage OTA in TSMC 180 nm technology

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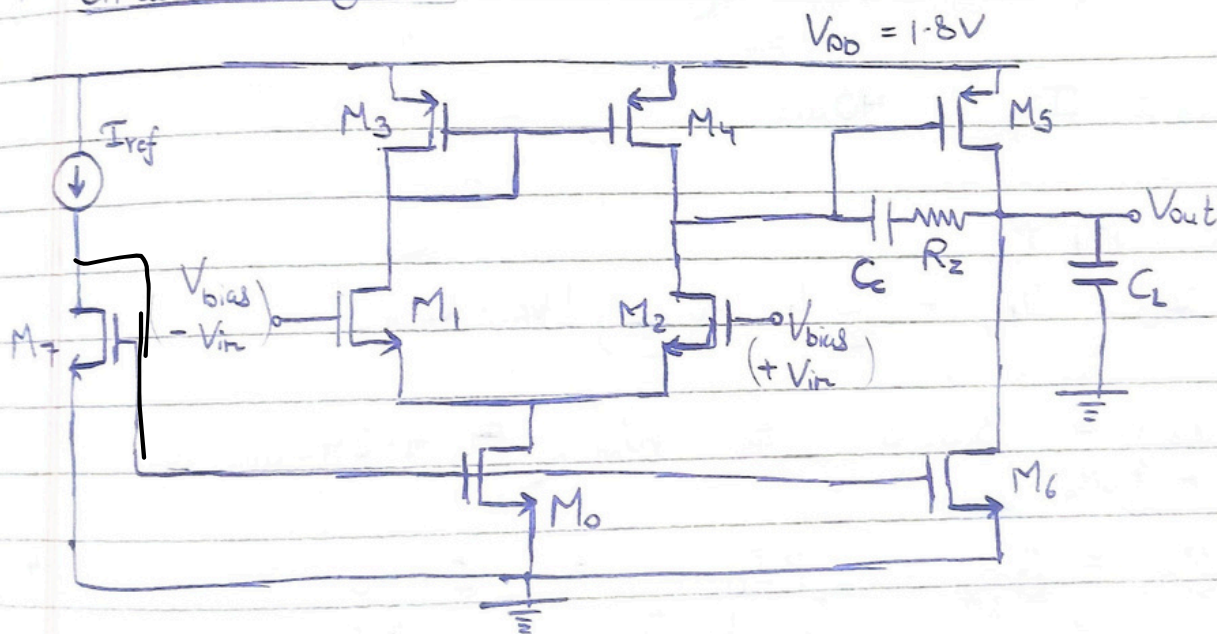
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Given: Model - TSMC 180 nm technology

$$V_{th,n} = 0.37V, V_{th,p} = 0.39V, \mu_n C_{ox} = 230 \mu A/V^2, \\ \mu_p C_{ox} = 100 \mu A/V^2, V_{dd} = 1.8V, L_{min} = 0.18 \mu m, W_{min} = 0.27 \mu m$$

Assume: $\lambda_n = \lambda_p = 0.1$, $C_L = 5pF$

* Circuit Diagram



Gate overdrive : $V_{as} - V_{th} = 0.2V$ (for all transistors)

Let $I_{ref} = 8\mu A$

For M_7 :

$$I_{ref} = I_D = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) [V_{as} - V_{th,n}]^2$$

$$\Rightarrow \left(\frac{W}{L} \right)_{M_7} = \frac{2 I_{ref}}{\mu_n C_{ox} (V_{as} - V_{th,n})^2} = \frac{2 \times 8 \times 10^{-6}}{230 \times 10^{-6} \times (0.2)^2} = 1.739$$

Let $L = 0.5 \mu m$ (for all mosfets)

$$g_{m7} = \frac{2I_D}{(V_{GS} - V_{th,n})} = \frac{2 \times 8 \times 10^{-6}}{0.2} = 8 \times 10^{-5} \Omega^{-1}$$

$$\therefore W_{M7} = 0.869 \mu m$$

Now, as the question states, current through M_0 (tail) should be 10 times of I_{ref} .

$$\therefore I_{M_0} = 80 \mu A$$

For M_0

$$I_{M_0} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_{th,n})^2$$

$$\Rightarrow \left(\frac{W}{L} \right)_{M_0} = 17.39 \Rightarrow W_{M_0} = 8.695 \mu m$$

Now, current flowing from M_1 & $M_2 = \frac{I_{M_0}}{2}$, as the circuit is symmetric.

$$\therefore I_{M_1} = I_{M_2} = 40 \mu A$$

For M_1, M_2

$$I_D = I_{M_1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) (V_{GS} - V_{th,n})^2$$

$$\therefore \left(\frac{W}{L} \right)_{M_1, M_2} = 8.695 \Rightarrow W_{M_1, M_2} = 4.348 \mu m$$

$$g_{m1} = g_{m2} = \frac{2I_D}{[V_{GS} - V_{th,n}]} = \frac{2 \times 40 \times 10^{-6}}{0.2} = 4 \times 10^{-4} \Omega^{-1}$$

$$r_{o1} = r_{o2} = \frac{1}{\lambda I_D} = 250 k\Omega$$

Now, M_1 is in saturation region,

$$\therefore V_{DS} \geq V_{GS} - V_{th,n} \quad - (1)$$

For max. output swing

$$V_G = V_{bias} = V_{DD}/2 = 0.9V$$

We also know for M_3 (PMOS) & M_4

$$V_{SG} - |V_{tp}| = 0.2V$$

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$$\Rightarrow (V_G)_{M3} = (V_G)_{M4} = 1.21V = (V_D)_{M1} \rightarrow \text{satisfies the above Condition (1)}$$

For $M3, M4$

$$I_D = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right) [V_{GS} - V_{tp}]^2$$

$$\Rightarrow \left(\frac{W}{L} \right)_{M3, M4} = \frac{2 \times 40 \times 10^{-6}}{100 \times 10^{-6} \times 0.04} = \underline{20} \Rightarrow W_{M3, M4} = 10 \mu m$$

$$r_{o3} = r_{o4} = \frac{1}{\lambda_p I_D} = 250 k\Omega$$

Gain of first stage : $A_{V1} = -g_{m1} (r_{o1} || r_{o3}) = \underline{-50}$

For $M6$

Let $I_D = 200 \mu A$.

Since the current in $M6$ is 25 times of current in $M7$, $\left(\frac{W}{L} \right)_{M6} = 25 \times \left(\frac{W}{L} \right)_{M7}$

$$\therefore \left(\frac{W}{L} \right)_{M6} = 43.475 \Rightarrow W_{M6} = 21.75 \mu m$$

For $M5$

$$g_{m5} = \frac{2 I_D}{(V_{SG} - |V_{tp}|)} = \frac{2 \times 200 \times 10^{-6}}{0.2} = 0.002$$

$$\left(\frac{W}{L} \right)_{M5} = \frac{2 I_D}{(V_{SG} - V_{tp})^2 \cdot \mu_p C_{ox}} = \frac{2 \times 200 \times 10^{-6}}{100 \times 10^{-6} \times (0.2)^2} = 100$$

$$\therefore W_{M5} = 50 \mu m$$

$$r_{o5} = r_{o6} = \frac{1}{\lambda I_D} = 50 k\Omega$$

Gain of second stage : $A_{V2} = -g_{m5} (r_{o5} || r_{o6}) = \underline{-50}$

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$$\text{Total Gain} = A_{V1} \cdot A_{V2} = 2500$$

Since, circuit is symmetric, $g_{m1} = g_{m2} = g_{m3} = g_{m4} = 4 \times 10^{-4} \text{ S}$
and $g_{m5} = g_{m6} = 0.002 \text{ S}$

~~and current is~~

and current in M_0 is twice that of M_1 ,

$$g_{m0} = 2g_{m1} = 8 \times 10^{-4} \text{ S}$$

* Poles & Zeros

$$\rightarrow P_1 = \frac{-1}{g_{m5}(r_{o1} \parallel r_{o3})(r_{o5} \parallel r_{o6})C_c}$$

(dominant)

$$\rightarrow P_2 = \frac{-g_{m5}}{C_L}$$

(non-dominant)

$$\rightarrow \text{Zero : } Z = \frac{g_{m5}}{C_c}$$

For phase margin close to 60° , $C_c \geq 0.22 C_L$

$$\therefore \text{Let } C_c = C_L/2 = 2.5 \text{ pF.}$$

$$\therefore P_1 = -64 \text{ K rad/s}$$

$$P_2 = -400 \text{ M rad/s}$$

$$Z = +800 \text{ M rad/s}$$

Now, to eliminate RHP zero, we add R_2 in series with C_c .

$$R_2 = 1/g_{m5} = 500 \Omega$$

$$\Rightarrow \text{A third pole is added, } P_3 = 1/R_2 C_c = -800 \text{ M rad/s}$$

$$\text{GBW Product : } \omega_{\text{unity}} = \frac{g_{m1}}{C_c} = -160 \text{ M rad/s}$$

$$\phi_M = 90 - \tan^{-1}\left(\frac{\omega_{\text{unity}}}{P_2}\right) - \tan^{-1}\left(\frac{\omega_{\text{unity}}}{P_3}\right)$$

$$\Rightarrow \phi_M = 56.89^\circ$$

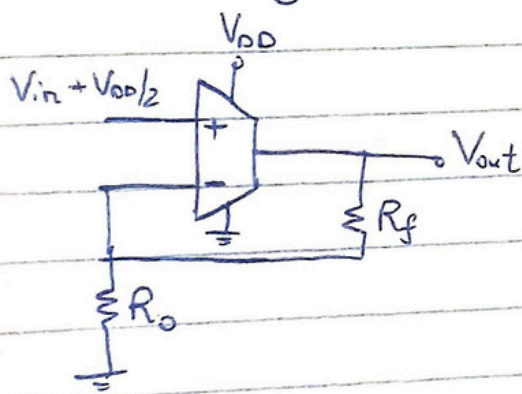
Power Consumption : $V_{DD} I_{total} = 1.8 (8 + 80 + 200) \mu W$
 $= 518.4 \mu W$

Voltage Swing limits

> Output limit : $V_{out} \geq 0.2 V$ & $V_{out} \leq 1.6 V$

> Input limit : $V_{in} \geq 0.53 V$ & $V_{in} \leq 1.15 V$

* For DC gain

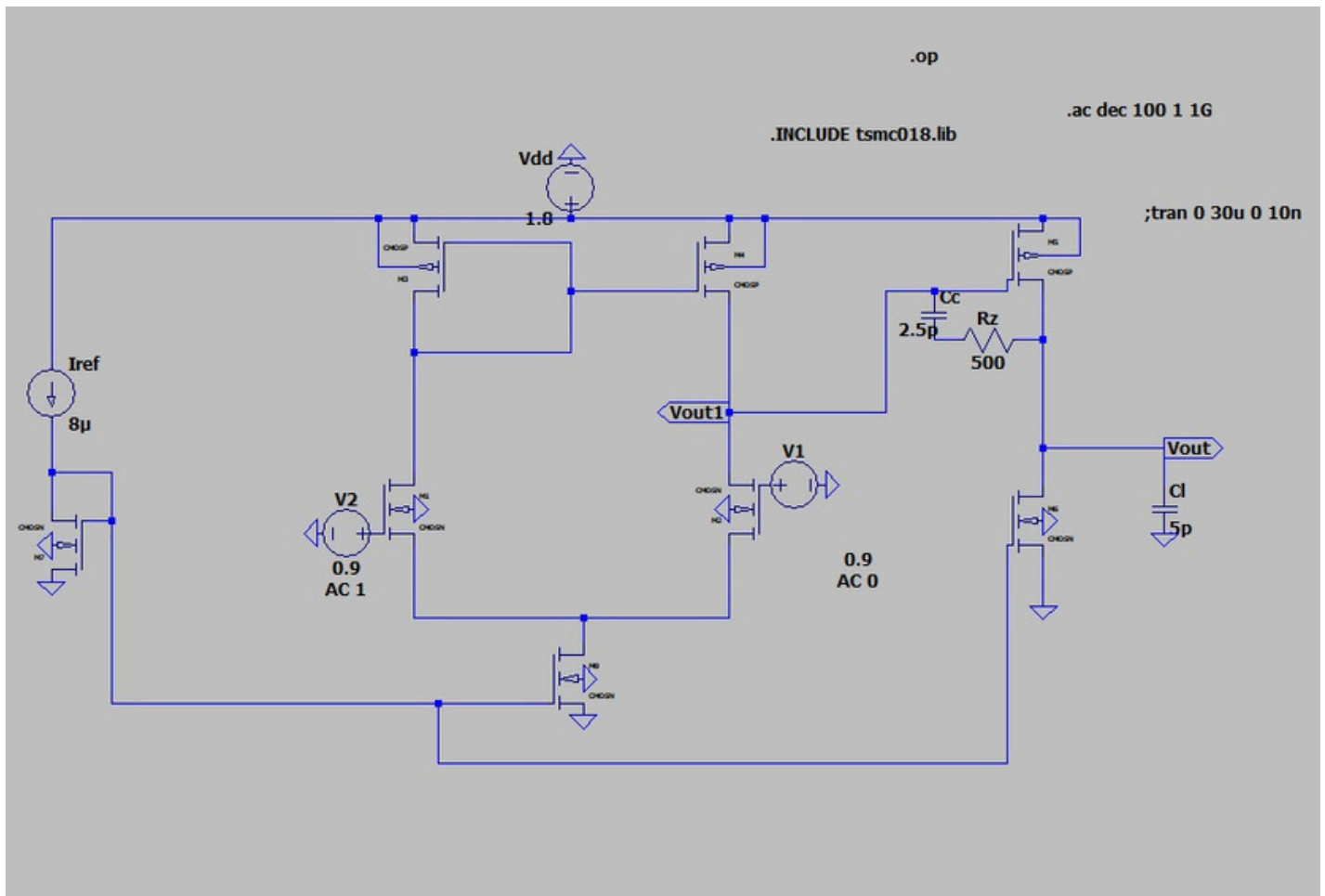


$$A = 1 + \frac{R_f}{R_o} = 2$$

$$\Rightarrow R_f = R_o = 1 M\Omega \text{ (let)}$$

LTSpice Simulation

Open-loop Circuit



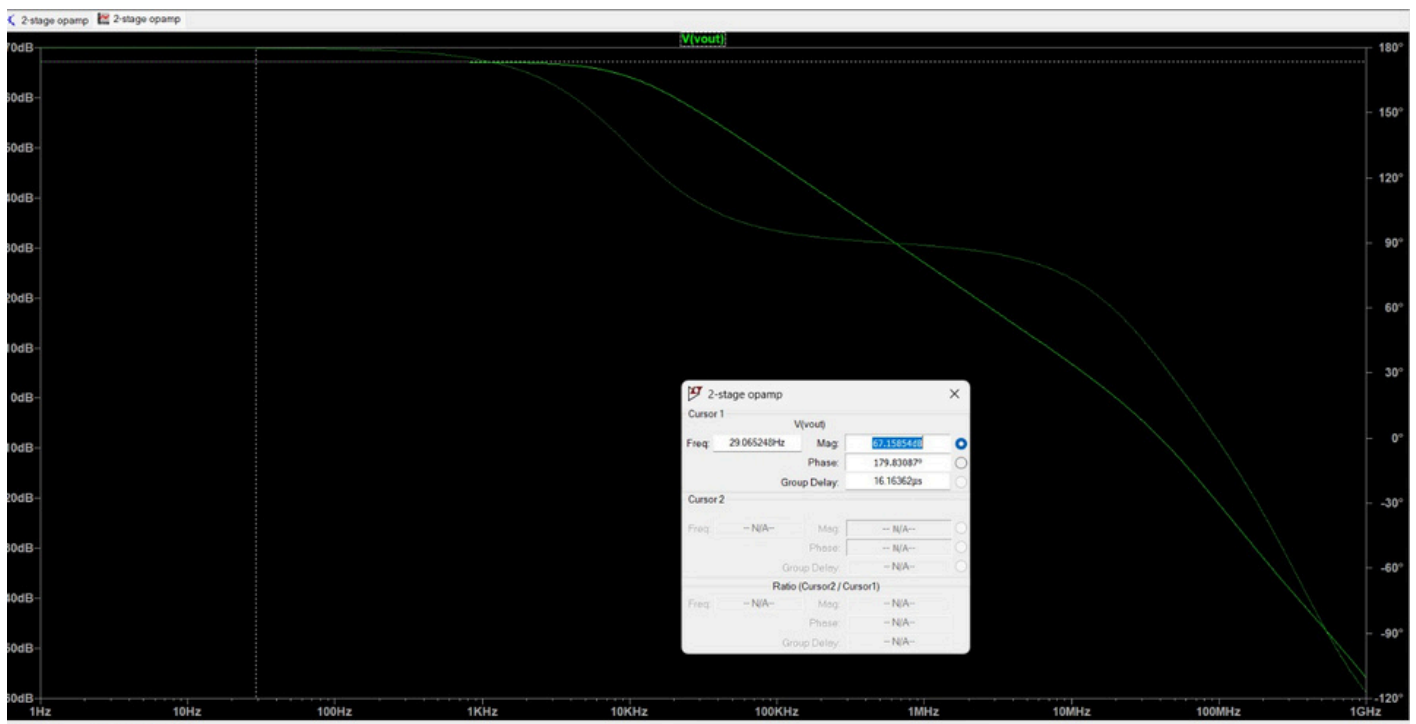
This is the open loop circuit with the differential inputs and all the conditions given in the problem.

DC Operating Point

Simulated values from LTspice (.op analysis) — TSMC 180 nm, $V_{DD} = 1.8\text{ V}$

MOSFET	Type	$ I_D $	g_m	r_o	$V_{GS} - V_{th}$
M0	NMOS	67.15 μA	671.45 μS	148.93 $\text{k}\Omega$	0.2 V
M1	NMOS	33.57 μA	335.73 μS	297.86 $\text{k}\Omega$	0.2 V
M2	NMOS	33.57 μA	335.73 μS	297.86 $\text{k}\Omega$	0.2 V
M3	PMOS	33.57 μA	335.73 μS	297.86 $\text{k}\Omega$	0.2 V
M4	PMOS	33.57 μA	335.73 μS	297.86 $\text{k}\Omega$	0.2 V
M5	PMOS	176.68 μA	1766.76 μS	56.60 $\text{k}\Omega$	0.2 V
M6	NMOS	176.68 μA	1766.76 μS	56.60 $\text{k}\Omega$	0.2 V
M7	NMOS	8.00 μA	80.00 μS	1.250 $\text{M}\Omega$	0.2 V

Note: $g_m = 2|I_D| / (V_{GS} - V_{th})$ | $r_o = 1 / (\lambda \cdot |I_D|)$ | $\lambda = 0.1\text{ V}^{-1}$ for both NMOS and PMOS | $V_{GS} - V_{th} = 0.2\text{ V}$ (design value)



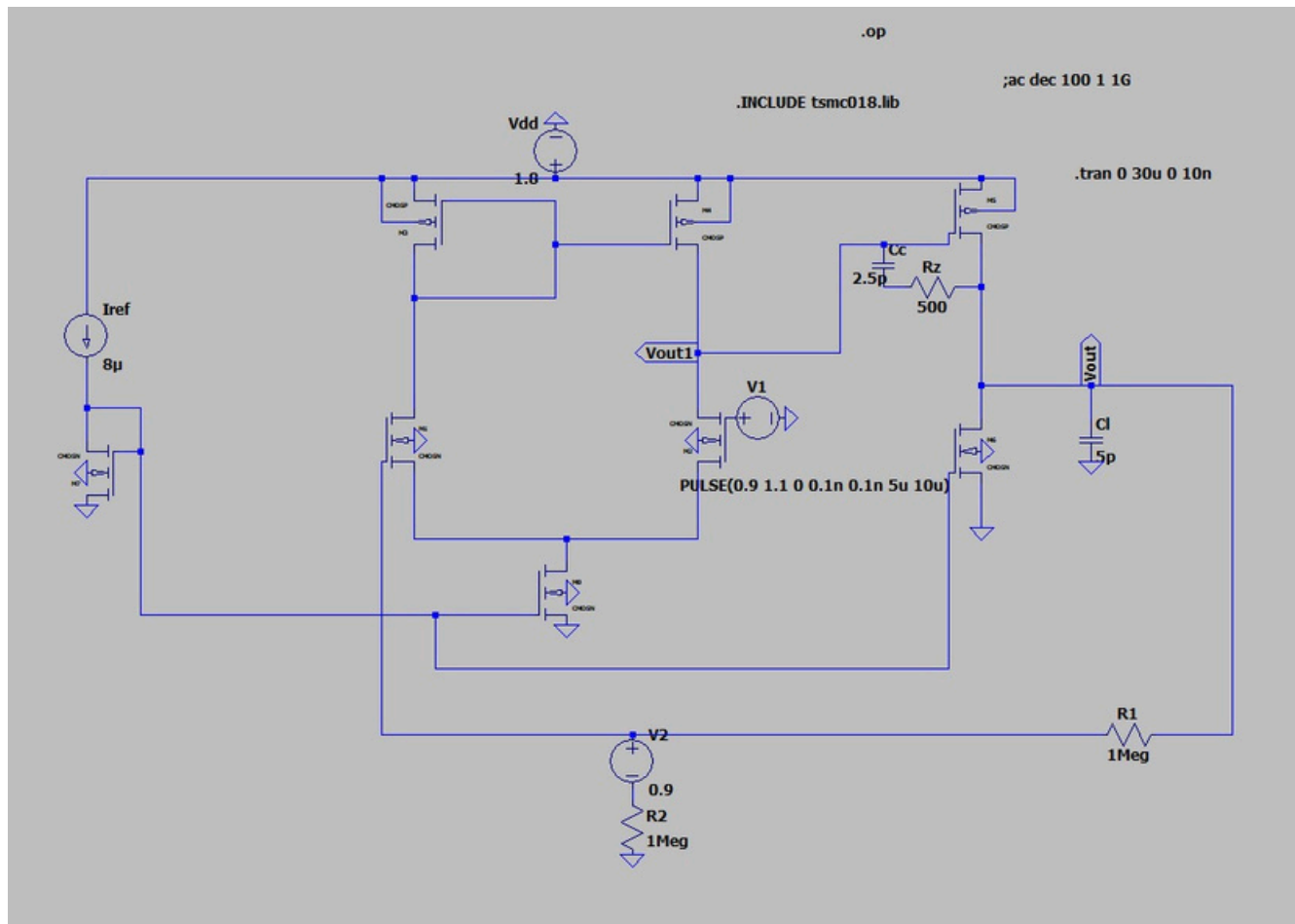
After running the simulation, we get an open loop gain of about 67.16dB, which is about 2300.



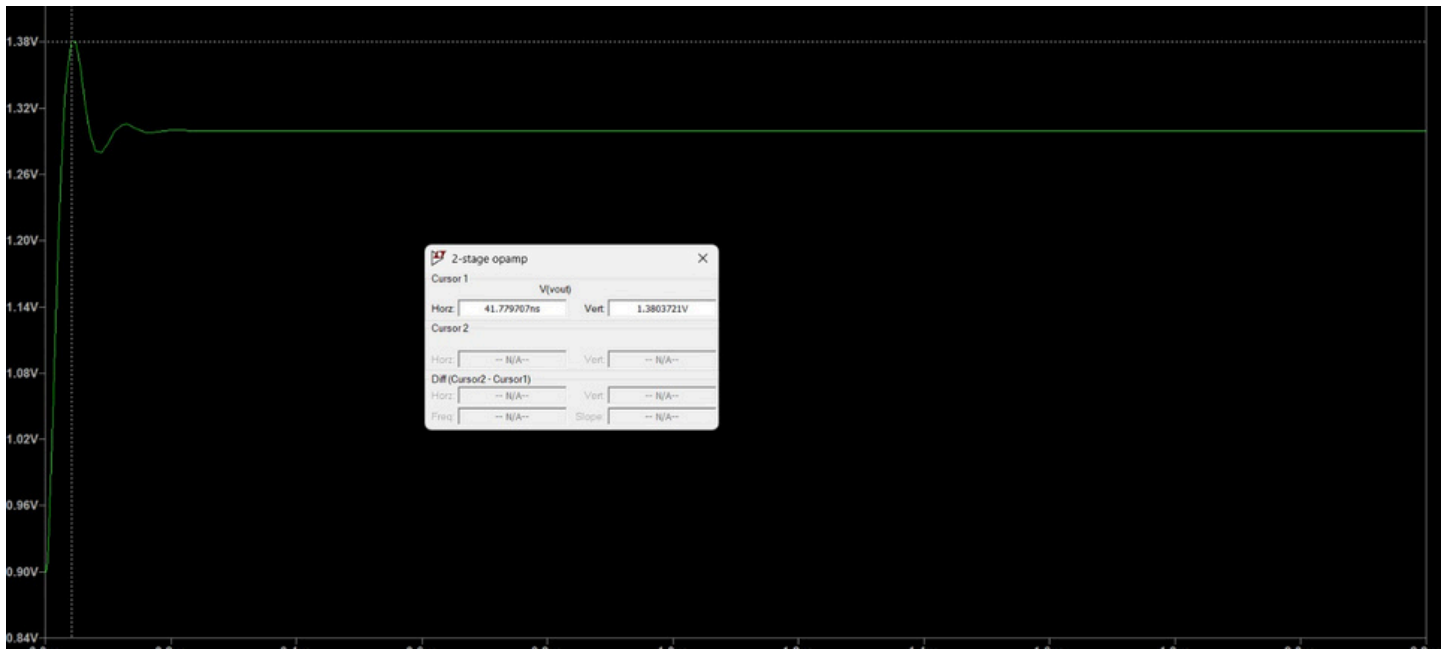
The gain crossover frequency (ω_{gc}) : $\omega(\text{unity}) = 20.04 \text{ MHz} = 125.91 \text{ Mrad/s}$.

The phase margin at $\omega_{gc} = 58.36 \text{ degrees}$.

Closed-loop Circuit



In the closed loop circuit, $R_1 = R_2 = 1\text{M ohm}$.
We have given 0.2V pulse step input, from 0.9V to 1.1V.



After running the simulation, we get a transient response starting from 0.9V and settling at 1.3V. Therefore, the DC gain of this circuit is 2.

Due to the presence of LHP poles in the system, we get an overshoot of about 6.15%.

Calculating the zeta from this gives 0.63, which aligns with our calculated open loop phase margin.